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Tadepalligudem

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Editor's Column

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Objective C

Objective-C is a general-purpose, object-oriented programming language that adds Smalltalk-style messaging to the C programming language. This is the main programming language used by Apple for the OS X and iOS operating systems and their respective APIs, Cocoa and Cocoa Touch. Objective-C is a thin layer atop C, and is a "strict superset" of C, means that it is possible to compile any C program with an Objective-C compiler, and to freely include C language code within an Objective-C class. Objective-C derives its object syntax from Smalltalk. All of the syntax for non-object-oriented operations (including primitive variables, pre-processing, expressions, function declarations, and function calls) are identical to those of C, while the syntax for object-oriented features is an implementation of Smalltalk-style messaging. The Objective-C model of object-oriented programming is based upon message passing to object instances. In Objective-C language one does not call a method one sends a message. This is unlike the Simula-style programming model used by C++. In Objective-C, the target of a message is resolved at runtime, with the receiving object itself interpreting the message. Method is identified by a selector or SEL a NULL-terminated string representing its name and resolved to a C method pointer implementing it: an IMP. A consequence of this is that the message-passing system has no type checking. The object to which the message is directed — the receiver — is not guaranteed to respond to a message, and if it does not, it raises an exception. Sending the message method to the object pointed to by the pointer obj would require the following code in C++:

```
obj->method(argument);
```

In Objective-C, this is written as follows:

```
[obj method:argument];
```

A Objective C program consists of various tokens and a token is either a keyword, an identifier, a constant, a string literal, or a symbol.



Foundation Framework:

Foundation Framework provides large set of features and they are listed below. It includes a list of extended datatypes like NSArray, NSDictionary, NSSet. It consists of a rich set of functions manipulating files, strings, etc. It provides features for URL handling, utilities like date formatting, data handling, error handling, etc.

A Objective-C program basically consists of the following parts:

Preprocessor Commands

Interface

Implementation

Methods

Variables

Statements & Expressions

Comments

Objective-C allows you to define protocols, which declare the methods expected to be used for a particular situation. Protocols are implemented in the classes conforming to the protocol.

If you need to add a method to an existing class, perhaps, to add functionality to make it easier to do something in your own application, the easiest way is to use a category.

The syntax to declare a category uses the @interface keyword, just like a standard Objective-C class description, but does not indicate any inheritance from a subclass. Instead, it specifies the name of the category in parentheses like this:

```
@interface ClassName (CategoryName)
```

```
@end
```

Objective-C permits a class to wholly replace another class within a program. The replacing class is said to "pose as" the target class.

For the versions that supported posing, all messages sent to the target class are instead received by the posing class.

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Salma Begum



TypeScript

TypeScript is a typed superset of JavaScript that compiles to plain JavaScript. TypeScript is pure object oriented with classes, interfaces and statically typed like C# or Java. The popular JavaScript framework Angular 2.0 is written in TypeScript. Mastering TypeScript can help programmers to write object-oriented programs and have them compiled to JavaScript, both on server side and client side.

By definition, "TypeScript is JavaScript for application-scale development." TypeScript is a strongly typed, object oriented, compiled language. It was designed by Anders Hejlsberg (designer of C#) at Microsoft. TypeScript is both a language and a set of tools. TypeScript is a typed superset of JavaScript compiled to JavaScript. In other words, TypeScript is JavaScript plus some additional features.

Features of TypeScript:

- *TypeScript is just JavaScript
- *TypeScript supports other JS libraries
- *JavaScript is TypeScript
- *TypeScript is portable

At its heart, TypeScript has the following three components

- *Language – It comprises of the syntax, keywords, and type annotations.
- *The TypeScript Compiler – The TypeScript compiler (tsc) converts the instructions written in TypeScript to its JavaScript equivalent.
- *The TypeScript Language Service – The "Language Service" exposes an additional layer around the core compiler pipeline that are editor-like applications. The language service supports the common set of a typical editor operations like statement completions, signature help, code formatting and outlining, colorization, etc.

When a TypeScript script gets compiled, there is an option to generate a declaration file (with the extension .d.ts) that functions as an interface to the components in the compiled JavaScript. The concept of declaration files is analogous to the concept of header files found in C/C++. The declaration files (files with .d.ts extension) provide intellisense for types, function calls, and variable support for JavaScript libraries like jQuery, MooTools, etc.

TypeScript is an Open Source technology. It can run on any browser, any host, and any OS. You will need the following tools to write and test a TypeScript program

A Text Editor

The text editor helps you to write your source code. Examples of a few editors include Windows Notepad, Notepad++, Emacs, vim or vi, etc. Editors used may vary with Operating Systems. The source files are typically named with the extension .ts

The TypeScript Compiler

The TypeScript compiler is itself a .ts file compiled down to JavaScript (.js) file. The TSC (TypeScript Compiler) is a source-to-source compiler (transcompiler / transpiler). The TSC generates a JavaScript version of the .ts file passed to it. In other words, the TSC produces an equivalent JavaScript source code from the TypeScript file given as an input to it. This process is termed as transpilation. However, the compiler rejects any raw JavaScript file passed to it. The compiler deals with only .ts or .d.ts files.

A TypeScript program is composed of

- *Modules
- *Functions
- *Variables
- *Statements and Expressions
- *Comments

An Example of typescript code:

```
var message:string = "Hello World"  
console.log(message)
```

When should I use TypeScript?

When your codebase is huge, and more than one person works on the project, a type system can help you avoid a lot of common errors. The TypeScript transpiler reveals the most obvious mistakes though it won't magically eliminate the need for debugging.

With TypeScript, we could eliminate a lot of type checks .

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A.Mahesh Kumar



Not only SQL

NoSQL Database is used to refer a non-SQL or non relational database. It provides a mechanism for storage and retrieval of data other than tabular relations model used in relational databases. NoSQL database doesn't use tables for storing data. It is generally used to store big data and real-time web applications.

History

In the early 1970, Flat File Systems are used. Data were stored in flat files and the biggest problems with flat files are each company implement their own flat files and there are no standards. It is very difficult to store data in the files, retrieve data from files because there is no standard way to store data. Then the relational database was created by E.F. Codd and these databases answered the question of having no standard way to store data. But later relational database also get a problem that it could not handle big data, due to this problem there was a need of database which can handle every types of problems then NoSQL database was developed.

Advantages of NoSQL

- *It supports query language.
- *It provides fast performance.
- *It provides horizontal scalability.

Types of NoSQL Databases

- *Document Oriented Databases – MongoDB, HBase, Cassandra, Amazon SimpleDB, Hypertable, etc.
- *Graph Based Databases – Neo4j, OrientDB, Facebook Open Graph etc.
- *Column Based Databases – CouchDB, OrientDB, etc.
- *Key Value Databases – Membase, Redis, MemcacheDB, etc.

NoSQL solutions choose to relax the notion of complete consistency to something called “eventual consistency.” This allows each system to make updates to data and learn of other updates made by other systems within a short period of time, without being totally consistent at all times.



RDBMS vs NOSQL

RDBMS

- Structured and organized data
- Structured query language (SQL)
- Data and its relationships are stored in separate tables.
- Data Manipulation Language, Data Definition Language
- Tight Consistency

NoSQL

- Stands for Not Only SQL
- No declarative query language
- No predefined schema
- Key-Value pair storage, Column Store, Document Store, Graph databases
- Eventual consistency rather ACID property
- Unstructured and unpredictable data
- CAP Theorem
- Prioritizes high performance, high availability and scalability
- BASE Transaction

A NoSql query consists of these parts:

- 1.fields to be extracted
- 2.table to extract the records from
- 3.expression for filtering the table rows
- 4.groupby - fields to group the data under
- 5.aggregate functions to be applied to columns in fields
- 6.orderby - fields to order the return data by
- 7.limit - an integer number of records to return.

NoSQL queries are constructed using JSON objects. Below is an example:

```
{ object: String, q: Expression,  
  fields: Array of String,  
  groupBy: Array of String,  
  aggregation: Object mapping fields to aggregate functions
```

} For example, the shortest query you can write would be: -15A81A0571

```
{ "object": "String", "q": "Expression" }
```

G.Sravya

GitHub

GitHub is a Web-based Git version control repository hosting service. It is mostly used for computer code. It offers all of the distributed version control and source code management (SCM) functionality of Git as well as adding its own features. It provides access control and several collaboration features such as bug tracking, feature requests, task management, and wikis for every project. GitHub offers both plans for private and free repositories on the same account which are commonly used to host open-source software projects. In addition to source code, GitHub supports the following formats and features:

- *Documentation, including automatically rendered README files in a variety of Markdown-like file formats (see README files on GitHub)
- *Issue tracking (including feature requests) with labels, milestones, assignees and a search engine
- *Wikis
- *Pull requests with code review and comments
- *Commits history
- *Graphs: pulse, contributors, commits, code frequency, punch card, network, members
- *Integrations Directory
- *Unified and split diffs
- *Email notifications
- *Option to subscribe someone to notifications by @ mentioning them.
- *GitHub Pages: small websites can be hosted from public repositories on GitHub. The URL format is <http://username.github.io>.
- *Nested task-lists within files
- *Visualization of geospatial data
- *3D render files that can be previewed using a new integrated STL file viewer that displays the files on a "3D canvas". The viewer is powered by WebGL and Three.js. Photoshop's native PSD format can be previewed and compared to previous versions of the same file.

PDF document viewer

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T.Sravani



Faculty Publications

Name of the Staff	Title of the Publication	Publication Details
Mrs.G.Loshma	Confidential Policy for Protecting User Images Uploaded on shared Websites	International journal of Scientific Research and Management(IJSRM). Vol 5, Issue 06 June 2017
Dr.V. S Naresh	An Efficient Privacy Preserving Technique Using Decoy Passwords	International journal of Scientific Research and Management(IJSRM). Vol 5, Issue 06 June 2017
Mr.V Venkateswara Rao	Hierarchical Clustering for Big Data Using Mapreduce in Hadoop Using Batch Updates	International Journal of Advanced Research in Computer Science and Software Engineering(IJARCSSE), Vol 7, Issue 6 June-17
	A Survey on Security Aspects of Server Virtualization in Cloud Computing	International Journal of Electrical and Computer Engineering (IJECE) Vol 7 No3 June 2017, pp-1326-133
Ms.D.Sasi Rekha	Secure Database as a Service for Cloud Tenants With no Exposure Of Unencrypted Data	International Journal of Science Engineering and Advance Technology (IJSEAT) Vol. 5 Issue 6, June -17
Dr.V. S Naresh	A Secure Anti-Collusion Data Sharing Scheme for Dynamic Groups in the Cloud by Using Identity based Encryption	International journal of Scientific Research and Management (IJSRM). Vol 5, Issue 06 June 2017
Mr.M R Raja Ramesh	Encryption of Data Using Radix-64 and Symmetric Key Encryption	International Journal of Advanced Research in Computer Science and Software Engineering(IJARCSSSE) Vol 7, Issue 6 June-17
Mrs.A.Leelavathi	Advanced demand forecasting for resource allocation in the cloud for media streaming Applications	International Journal for Modern trends in Science and Technology ISSN:2455-3778,Vol 3,Issue 9,September 2017

PLACEMENTS INFORMATION

S.NO	ROLLNO	NAME OF THE STUDENT	COMPANY
1	13A81A0568	CHILAKAPATI RAMCHARAN SRIHARSHA	TCS,JUSPAY Technologies Pvt ltd
2	14A81A0517	GHANTASALA TANUJA	
3	14A81A05G8	SAI RAJESH UPPULURI	JUSPAY Technologies Pvt ltd
4	14A81A05E6	KUNAPAREDDY PAVANI KUMARI	Technovert Solutions
5	14A81A0577	KALLA RAJA GANGADHAR	
6	14A81A0569	DANDAMUDI MADHAVI	
7	14A81A05B7	VEMPALI RAVI KUMAR	

DETAILS OF GUESTLECTURES

SNo	Title of Guest Lecture	Name of Resource Person, Organization Name	Date	Target Audience
1.	An Insight to Bigdata	Dr J.V.R Murthy, Director Jntuk	13-09-2017	III B.Tech CSE
2.	Bossmool Orientation Programme	Prof .D.Janaki Ram,IIT Karaghpur Prof. D.V.L.N Somayajulu,NIT Warangal	14-10-2017	III B.Tech CSE

On the eve of Engineer's Day various technical events have been conducted in the department. The list entitled as:

S.No.	Event Name	Roll No	Student Name	Rank
1	Logo Designing	15A81A0550	V. Ramalakshmi	1 st Prize
		16A81A0561	A. Sai Lakshmi	2 nd Prize
2	Technical Quiz	15A81A0586	M. Jayanth	1 st Prize
		15A81A0588	M. Purna Vamsi	
		15A81A0596	P. Manideep	
		14A81A0575	K.V.G. Satish	2 nd Prize
		14A81A0576	K. Vamsi Krishna	
		14A81A05A6	P.S.G. Sri Ram	
3	Tech Talk	16A81A0516	G. Chaitanya	1 st Prize
		16A81A05N0	T. Neelima	2 nd Prize
4	Debate	14A81A0514	G. Mounika	1 st Prize
		14A81A05G2	P. Manjari	
		16A81A0508	Ch. Sudeepa	2 nd Prize
		15A81A05K3	K. Sri Valli	
5	Image Interpretation	14A81A0531	K. Vijaya Lakshmi	1 st Prize
		16A81A05N0	T. Neelima	2 nd Prize
6	Chat Bot (Static)	16A81A0516	G.S.K. Chaitanya	1 st Prize
		14A81A0539	N. Divya	2 nd Prize
7	Debugging	16A81A0590	J. Sailaja	1 st Prize
		15A81A05F1	M.L. Mounika	2 nd Prize
8	Treasure Hunt	14A81A0541	N. Mohan Krishna	Winners
		14A81A05B2	S. Alekhya	
		15A81A0530	M. Supraja	
		15A81A0588	M. Vamsi	
		14A81A05J7	K. Satya Sri	
		14A81A05L0	N. Devi	
		16A81A0561	A. Sai Lakshmi	

A GUEST LECTURE ON AN INSIGHT TO BIGDATA



ENGINEER'S DAY CELEBRATIONS



FRESHER'S DAY CELEBRATIONS



SAHAYA ACTIVITY

